

SYSTEMS ANALYSIS

Systems Analysis & Design

Author: Eng. Carlos Andrés Sierra, M.Sc.
cavirguezs@udistrital.edu.co

Full-time Adjunct Professor
Computer Engineering Program
School of Engineering
Universidad Distrital Francisco José de Caldas

2025-III



UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Outline

- 1 Basic Concepts
- 2 Chaos and Dynamic Systems
- 3 Abstraction and Modularity



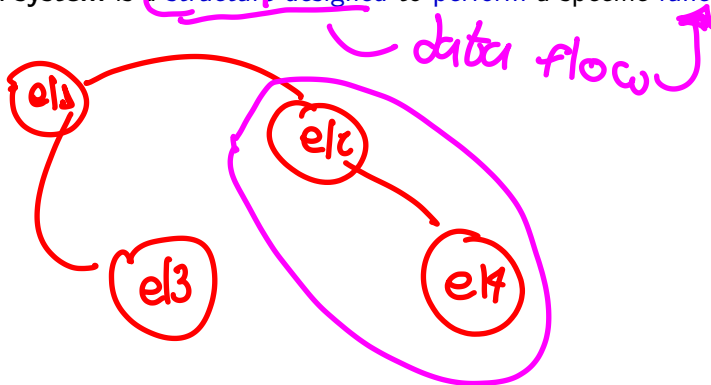
Outline

- 1 Basic Concepts
- 2 Chaos and Dynamic Systems
- 3 Abstraction and Modularity



What is a System?

- A **system** is a set of interacting components that work together to achieve a common goal.
- A **system** is a collection of elements organized in a specific way.
- A **system** is a structure designed to perform a specific function.



Systems Analysis Process

- **Systems analysis** is the *process* of *studying* a *system* in order to identify its *components*, *interactions*, and *goals*.
- *Systems analysis* is the *process* of understanding how a system works and how it can be improved.
- *Systems analysis* is the first step in the systems development lifecycle.



Systems Analysis Process

- **Systems analysis** is the *process* of **studying** a **system** in order to identify its **components**, **interactions**, and **goals**.
- **Systems analysis** is the *process* of **understanding** how a **system works** and how it can be **improved**.
- **Systems analysis** is the *first step* in the systems development lifecycle.

Optimization
↓
engineering



Systems Analysis Process

- **Systems analysis** is the *process* of **studying** a **system** in order to identify its **components**, **interactions**, and **goals**.
- **Systems analysis** is the *process* of **understanding** how a **system works** and how it can be **improved**.
- **Systems analysis** is the first step in the **systems development lifecycle**.

SDLC

requirements



Systems Development Lifecycle

- The **systems development lifecycle** is a *process* that *guides* the development of a system
- It includes **planning**, **analysis**, **design**, **implementation**, and **maintenance** phases.
- It is a **structured approach** to *developing* a system that ensures it meets the needs of its users.

step by step



Systems Development Lifecycle

- The **systems development lifecycle** is a *process* that *guides* the *development* of a *system*.
- It includes *planning*, *analysis*, *design*, *implementation*, and *maintenance* phases.
- It is a **(structured approach)** to *developing* a *system* that *ensures* it meets the *needs* of its *users*.



- final
user

(fix)
client & final user

quickly

Questions

Warehouse

web (1997)

100
3 per cent

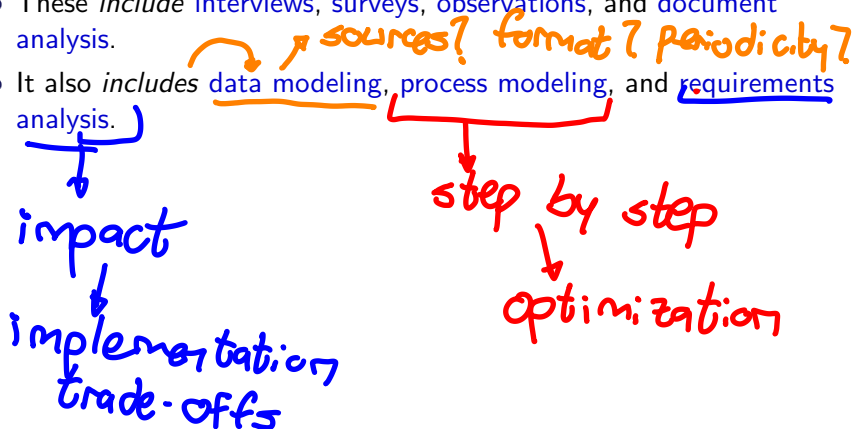
information gathering

Block Buster



Systems Analysis Techniques

- **Systems analysis** uses a *variety* of **techniques** to *study* a **system**.
- These *include* **interviews**, **surveys**, **observations**, and **document analysis**.
- It also *includes* **data modeling**, **process modeling**, and **requirements analysis**.



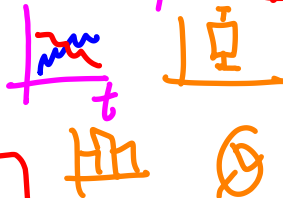
Systems Analysis Tools

- **Systems analysis** uses a variety of tools to study a system.
- These include **diagrams**, charts, flowcharts, and **data models**.
- It also includes software tools such as spreadsheets, databases, and simulation software.

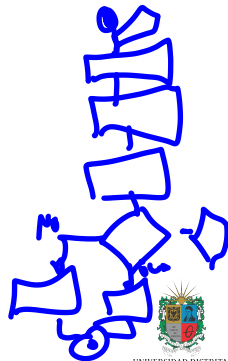
DB
Pipeline
ETL



Describe
Analysis

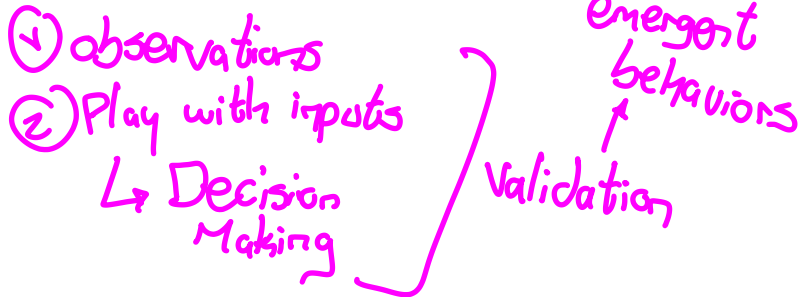


Process



Systems Analysis Tools

- **Systems analysis** uses a *variety* of **tools** to *study* a **system**.
- These *include* **diagrams**, **charts**, **flowcharts**, and **data models**.
- It also *includes* software tools such as (spreadsheets), databases, and simulation software.



Lateral Thinking

- **Lateral thinking** is a creative problem-solving technique that involves thinking outside the box.
 - It is a non-linear approach to problem-solving that encourages innovation and creativity.
 - It is a useful technique for generating new ideas and solving complex problems.
 - Examples:
 - How can you improve the design of a product?
 - What are the benefits of failure?
 - Why is ignorance important?
 - When is failure better than success?
- tech → startups

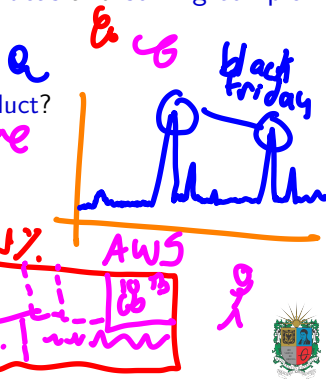


Lateral Thinking

- **Lateral thinking** is a creative problem-solving technique that involves thinking outside the box.
- It is a non-linear approach to problem-solving that encourages innovation and creativity.
- It is a useful technique for generating new ideas and solving complex problems.
- Examples:

- How can you improve the design of a product?
- What are the benefits of failure?
- Why is ignorance important? → learn
- When is failure better than success?

business opportunity



Lateral Thinking Training

- **Lateral thinking** is a *skill* that can be learned and developed through training and practice.
- It involves exercises, games, and activities that encourage creative thinking.
- Examples of lateral thinking exercises:
 - Brainstorming sessions
 - Mind mapping exercises.
 - Role-playing games.
 - Problem-solving activities.

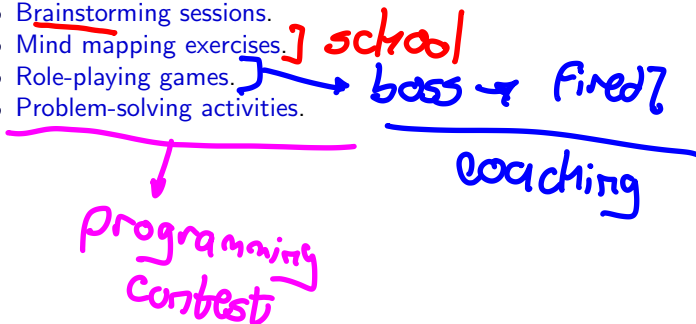
brain storm

go
chess



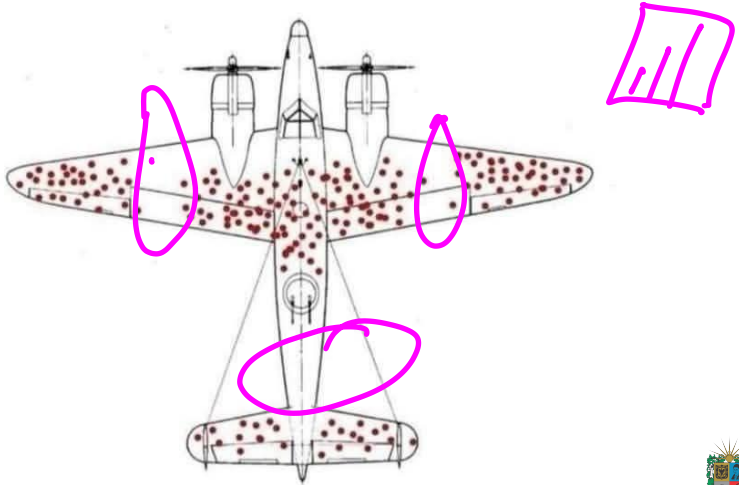
Lateral Thinking Training

- **Lateral thinking** is a *skill* that can be **learned** and **developed** through training and practice.
- It involves **exercises**, **games**, and **activities** that encourage creative thinking.
- Examples of lateral thinking exercises:
 - Brainstorming sessions.
 - Mind mapping exercises.
 - Role-playing games.
 - Problem-solving activities.



Lateral Thinking Exercise

World War II, the perspective of Abraham Wald:



Uncertainty and Risk

- **Uncertainty** is the lack of knowledge about the future outcome of a decision or event.

- Risk is the probability of a negative outcome or loss associated with a decision or event.

- Uncertainty and risk are inherent in complex systems and decisions.

- They can be managed through planning, analysis, and mitigation strategies.

+ information →
 + knowledge
 X simulation
 Probability



Uncertainty and Risk

- **Uncertainty** is the lack of knowledge about the future outcome of a decision or event.
 - **Risk** is the probability of a negative outcome or loss associated with a decision or event.
 - Uncertainty and risk are inherent in complex systems and decisions.
 - They can be managed through planning, analysis, and mitigation strategies.
- caso 170!*



Uncertainty and Risk

- **Uncertainty** is the **lack** of **knowledge** about the **future outcome** of a decision or event.
- **Risk** is the **probability** of a **negative outcome** or **loss** associated with a decision or event.
- **Uncertainty** and **risk** are *inherent* in **complex systems** and **decisions**.
- They can be managed through **planning**, **analysis**, and **mitigation strategies**.



Uncertainty and Risk

- **Uncertainty** is the **lack** of **knowledge** about the **future outcome** of a decision or event.
- **Risk** is the **probability** of a **negative outcome** or **loss** associated with a decision or event.
- **Uncertainty** and **risk** are *inherent* in **complex systems** and **decisions**.
- They can be managed through **planning**, **analysis**, and **mitigation strategies**.

③

software
projects

①

②

agile



Complexity and Emergence

- **Complexity** is the degree to which a system is difficult to understand.

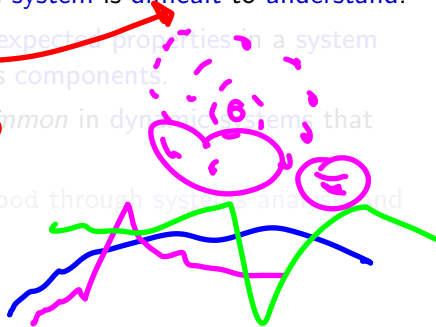
- Emergence is the appearance of unexpected properties in a system that arise from the interactions of its components.

- Complexity is more common in dynamic systems that are non-linear and chaotic.

- They can be studied and understood through systems analysis and modeling.

(+ elements
x relations)

Planet Nine



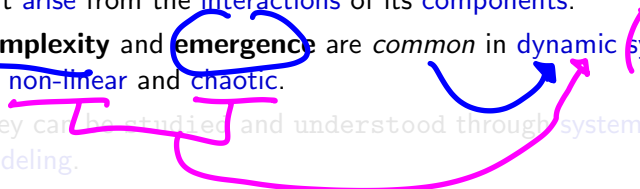
Complexity and Emergence

- **Complexity** is the degree to which a system is difficult to understand.
- **Emergence** is the appearance of (unexpected properties) in a system that arise from the interactions of its components.

- Complexity and emergence are common in dynamic systems that are non-linear and chaotic.
- They can be studied and understood through systems analysis and modeling.



Complexity and Emergence

- **Complexity** is the degree to which a system is difficult to understand.
 - **Emergence** is the appearance of unexpected properties in a system that arise from the interactions of its components.
 - **Complexity** and **emergence** are common in dynamic systems that are non-linear and chaotic.
 - They can be studied and understood through systems analysis and modeling.
- 



Complexity and Emergence

- **Complexity** is the degree to which a system is difficult to understand.
- **Emergence** is the appearance of unexpected properties in a system that arise from the interactions of its components.
- **Complexity** and **emergence** are common in dynamic systems that are non-linear and chaotic.
- They can be studied and understood through systems analysis and modeling.

→ Exploratory Data Analysis
 ↳ ydata-profiling
 → correlation matrix
 ↳ kaggle



Outline

- 1 Basic Concepts
- 2 Chaos and Dynamic Systems
- 3 Abstraction and Modularity



What is Chaos?

- **Chaos** is a *branch* of **mathematics** that *studies* the **sensitivity** of **dynamical systems** to **initial conditions**.
- **Chaos** is a **non-linear** behavior that is **highly sensitive** to **initial conditions**.
- **Chaos** is a **deterministic** behavior that is **not predictable** in the **long term**.
- **Chaos** is a **complex** behavior that is **hard** to **understand**.

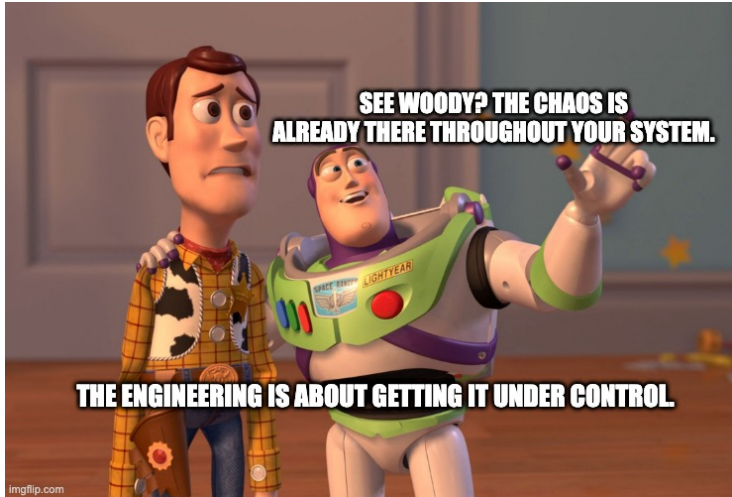


What is Chaos?

- **Chaos** is a *branch* of **mathematics** that *studies* the **sensitivity** of **dynamical systems** to **initial conditions**.
- **Chaos** is a **non-linear** behavior that is **highly sensitive** to **initial conditions**.
- **Chaos** is a **deterministic** behavior that is **not predictable** in the **long term**.
- **Chaos** is a **complex** behavior that is **hard** to **understand**.



Chaos is Everywhere!



What is a Dynamic System?

- A **dynamic system** is a system that **changes** over time.
- A **dynamic system** is **sensitive** to **initial conditions**.
- A **dynamic system** is **non-linear**.
- A **dynamic system** can be **chaotic**.



What is a Dynamic System?

- A **dynamic system** is a system that **changes** over time.
- A **dynamic system** is **sensitive** to **initial conditions**.
- A **dynamic system** is **non-linear**.
- A **dynamic system** can be **chaotic**.



Chaotic Attractors

A **chaotic attractor** is a set of points in a phase space that attracts the trajectories of a dynamical system.



Fractals in Nature



Watch this video: <https://www.youtube.com/watch?v=kkGeOWYOFoA>



Swarm Intelligence I

- **Swarm intelligence** is the collective behavior of *decentralized, self-organized systems*, natural or artificial.
- The concept is employed in work on **artificial intelligence**.
- The expression was introduced by *Gerardo Beni* and *Jing Wang* in 1989, in the context of **cellular robotic systems**. For example, watch this **video**.



Swarm Intelligence I

- **Swarm intelligence** is the collective behavior of *decentralized, self-organized systems*, natural or artificial.
- The concept is employed in work on **artificial intelligence**.
- The expression was introduced by *Gerardo Beni* and *Jing Wang* in **1989**, in the context of **cellular robotic systems**. For example, watch this **video**.



Swarm Intelligence II

- The **idea** is: if you see an **individual**, it may seem random; **however**, several **individuals interacting** with each other and the environment show **smart behaviors**.
- Yu Takeuchi said: one colombian is more intelligent than one japanese, but two japanese are smarter than two colombians.
- There are interesting **population behaviors** in nature, specially in **insects**: bees, ants, termites, among others.
- There are also many **examples in nature**: schools of fish, birds, wolves.



Swarm Intelligence II

- The **idea** is: if you see an **individual**, it may seem random; **however**, several **individuals interacting** with each other and the environment show **smart behaviors**.
- **Yu Takeuchi** said: **one colombian** is **more** intelligent than **one japanese**, but **two japanese** are **smarter** than **two colombians**.
- There are interesting **population behaviors** in nature, specially in **insects**: bees, ants, termites, among others.
- There are also many **examples in nature**: schools of fish, birds, wolves.



Swarm Intelligence II

- The **idea** is: if you see an **individual**, it may seem random; **however**, several **individuals interacting** with each other and the environment show **smart behaviors**.
- **Yu Takeuchi** said: **one colombian** is **more** intelligent than **one japanese**, but **two japanese** are **smarter** than **two colombians**.
- There are interesting **population behaviors** in nature, specially in **insects**: bees, ants, termites, among others.
- There are also many **examples** in **nature**: schools of fish, birds, wolves.

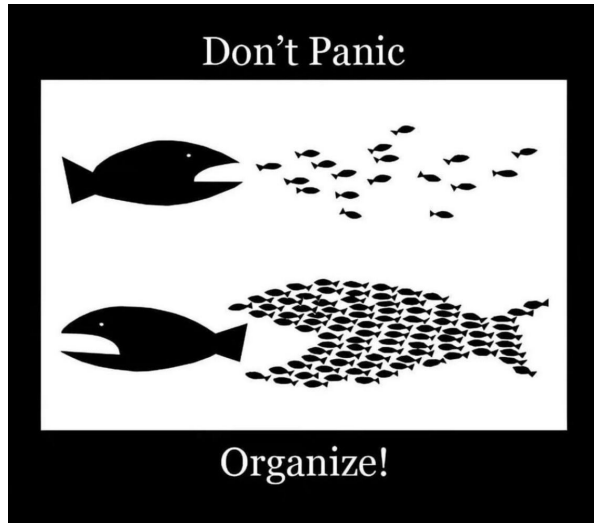


Emergent Behaviors

- **Emergent behavior** is the **appearance** of **complex patterns** and behaviors from **many** simple interactions.
- **Emergent behavior results** from the **collective** behavior of the **individuals** in the system.
- **Emergent behavior** is **not planned** or **designed** by any individual, but **arises** from their **interactions**.
- **Emergent behavior** is **not** the **sum** of the **individual** behaviors, but **something more**: **synergy**.
- **Swarm intelligence** refers to interesting **emergent behaviors**.



School Fish Algorithm



School Fish Algorithm

- **School fish** are interesting. When a predator attacks, they become confused by the large number of individuals and their **diverse movements**.
- The **idea** is simple: *"Don't touch me, don't come too close, but stay somewhat close."*
- This behavior is a **chain of action and reaction**. It **confuses predators** and helps the school move uniformly.
- Do you remember **Nemo**? The fish with a sword snout, the pirates, or Marlin's imitation of talking-all are somewhat similar. Watch **here**.
- The **school fish algorithm** is a **multi-agent system** that **simulates** the behavior of a **school of fish**.



School Fish Algorithm

- **School fish** are interesting. When a predator attacks, they become confused by the large number of individuals and their **diverse movements**.
- The **idea** is simple: *"Don't touch me, don't come too close, but stay somewhat close."*
- This behavior is a **chain of action and reaction**. It **confuses predators** and helps the school move uniformly.
- Do you remember **Nemo**? The fish with a sword snout, the pirates, or Marlin's imitation of talking-all are somewhat similar. Watch **here**.
- The **school fish algorithm** is a **multi-agent system** that **simulates** the behavior of a **school of fish**.



School Fish Algorithm

- **School fish** are interesting. When a predator attacks, they become confused by the large number of individuals and their **diverse movements**.
- The **idea** is simple: *“Don’t touch me, don’t come too close, but stay somewhat close.”*
- This behavior is a **chain of action and reaction**. It **confuses predators** and helps the school move uniformly.
- Do you remember **Nemo**? The fish with a sword snout, the pirates, or Marlin’s imitation of talking-all are somewhat similar. Watch **here**.
- The **school fish algorithm** is a **multi-agent system** that **simulates** the behavior of a **school of fish**.



Ant Colony Algorithm

- Ant colony algorithm is a **multi-agent system** that **simulates** the behavior of an **ant colony**.
- Ant colony algorithm is based on the **social behavior** of **ants** and the use of **pheromones**. Watch **here**.
- Ant colony algorithm is used to solve optimization problems. Watch **here**.



Ant Colony Algorithm

- **Ant colony algorithm** is a **multi-agent system** that **simulates** the behavior of an **ant colony**.
- **Ant colony algorithm** is based on the **social behavior** of **ants** and the use of **pheromones**. Watch **here**.
- **Ant colony algorithm** is used to **solve optimization problems**. Watch **here**.



Outline

- 1 Basic Concepts
- 2 Chaos and Dynamic Systems
- 3 Abstraction and Modularity



What is Abstraction?

- **Abstraction** is the *process* of **ignoring** minor **details** in order to **focus** on the **important aspects** of a **system**.
- **Abstraction** is the *process* of **simplifying** a complex system to **understand** it.
- **Abstraction** is the *process* of **generalizing** a specific system to **apply** it to **other systems**.



What is Abstraction?

- **Abstraction** is the *process* of **ignoring** minor **details** in order to **focus** on the **important aspects** of a **system**.
- **Abstraction** is the *process* of **simplifying** a **complex system** to **understand** it.
- **Abstraction** is the *process* of **generalizing** a **specific system** to **apply** it to **other systems**.



What is Abstraction?

- **Abstraction** is the *process* of **ignoring** minor **details** in order to **focus** on the **important aspects** of a **system**.
- **Abstraction** is the *process* of **simplifying** a **complex system** to **understand** it.
- **Abstraction** is the *process* of **generalizing** a **specific system** to **apply** it to **other systems**.



What is Modularity?

- **Modularity** is the *process* of **dividing** a **system** into **smaller parts** called **modules**.
- **Modularity** is the *process* of **organizing** a **system** into **independent units** that can be **developed** and **maintained** separately.
- **Modularity** is the *process* of **reducing** the **complexity** of a **system** by **breaking** it into **smaller parts**.



What is Modularity?

- **Modularity** is the *process* of **dividing** a **system** into **smaller parts** called **modules**.
- **Modularity** is the *process* of **organizing** a **system** into **independent units** that can be **developed** and **maintained separately**.
- **Modularity** is the *process* of **reducing** the **complexity** of a **system** by **breaking** it into **smaller parts**.



What is Modularity?

- **Modularity** is the *process* of **dividing** a **system** into **smaller parts** called **modules**.
- **Modularity** is the *process* of **organizing** a **system** into **independent units** that can be **developed** and **maintained separately**.
- **Modularity** is the *process* of **reducing** the **complexity** of a **system** by **breaking** it into **smaller parts**.



Abstraction and Modularity

- **Abstraction** and **modularity** are *two important concepts* in **systems analysis**.
- They help **reduce** the **complexity** of a **system** by **ignoring details** and **dividing** it into **smaller parts**.
- They help **improve** the **understanding**, **development**, and **maintenance** of a **system**.



Abstraction and Modularity

- **Abstraction** and **modularity** are *two important concepts* in **systems analysis**.
- They help **reduce** the **complexity** of a **system** by **ignoring details** and **dividing** it into **smaller parts**.
- They help **improve** the **understanding**, **development**, and **maintenance** of a **system**.



Outline

- 1 Basic Concepts
- 2 Chaos and Dynamic Systems
- 3 Abstraction and Modularity



Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

